Unlock the Lab: sample studies

Six studies for discussion, with an answer key for the facilitator

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About this set

These six fictional studies come from Unlock the Lab and are printed as participants see them in the app, with the two answers that the app asks for. Use them for discussion in groups, as a warm-up before the app or for anyone taking part on paper. The evaluation guide has the rubric and the glossary to go with them.

The set covers the three quality bands of the expert ratings: Low quality (expert rating 1–2), Medium quality (expert rating 3–5) and High quality (expert rating 6–7). From each band it takes the study that is hardest to judge, with the most criteria whose band differs from the overall one, and the clearest, with the fewest, where possible with a different expert rating. The studies keep their order in the app, so their letters say nothing about their quality.

Each study is on its own page. The answer key at the end gives the expert's rating, a summary and a note on each criterion; keep it back until everyone has answered. To play 'Predict the crowd' on paper, collect the ratings, work out the group's average for each study and compare it with the predictions. The live dashboard shows how everyone who has used the app rated each study, under the dashboard number given in the answer key.

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Study A · Fictional study

TITLE

The neuroprotective effects of a daily herbal tea infusion on Alzheimer's disease prevention

Access

Free to read online (Open Access). As with many open-access journals, authors pay a publication fee so that readers do not have to.

Source

Predatory Journal. The journal website contains broken English and promises 'publication within 48 hours for a fee', indicating zero peer review.

Theory

The premise is loosely based on traditional folk medicine practices. It ignores current neurological understanding of amyloid plaques and tau tangles, which are the primary biological markers of Alzheimer's disease.

Methods & Data

A 6-week observational study of 40 people who already chose to drink the tea daily. There was no randomised control group. Cognition was measured via brief phone interviews rather than validated in-person cognitive assessments. **Data availability:** 'Raw data unavailable due to participant confidentiality.'

Conclusion

Overblown. The study confuses a short-term feeling of alertness (likely an expectation or placebo effect) with the long-term prevention of a complex neurodegenerative disease.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study B · Fictional study

TITLE

A causal link between Instagram use and adolescent anxiety

Access

Accessible with difficulty (via university library subscription).

Source

Mid-Tier Psychology Journal. A legitimate peer-reviewed journal, though one that occasionally publishes correlational studies with catchy titles to boost its impact factor.

Theory

Uses standard social psychology theories regarding 'social comparison' and self-esteem. However, it assumes a direct causal vector without accounting for the complexity of adolescent mental health.

Methods & Data

A cross-sectional survey of 5,000 teenagers taken at a single point in time. While the sample size is large, the method relies entirely on self-reporting for both 'hours used' and 'anxiety levels'. **Data availability:** Summary statistics provided; raw data available only upon request to the lead author.

Conclusion

Logical but limited. The study establishes a strong correlation but the authors (and the title) imply causation. They fail to address whether anxious teens simply retreat to social media more often.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study C · Fictional study

TITLE

An evolutionary explanation for the correlation between height and CEO status in males

Access

Freely available for download (preprint server). Because the paper has not been accepted by a journal, there is no paywall — anyone can read it immediately.

Source

Preprint Server. The paper has been uploaded to a server but has not yet been peer-reviewed or accepted by a journal.

Theory

Uses evolutionary psychology to suggest height signals dominance. It fails to account for the 'halo effect' (social bias) and treats modern corporate structures as if they are prehistoric hunter-gatherer societies.

Methods & Data

The authors looked at the height of CEOs of Fortune 500 companies (the 500 largest US firms) and compared them to the national average. No measure of actual company performance, profit or leadership skill was included. **Data availability:** A spreadsheet of CEO heights is available on request.

Conclusion

Overblown. It confuses a societal bias (we promote tall men) with an inherent evolutionary ability (tall men are better leaders).

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study D · Fictional study

TITLE

The efficacy of mindfulness meditation as an intervention for chronic pain: A systematic review

Access

Open Access via Cochrane Library.

Source

Cochrane Database of Systematic Reviews. Gold-standard source for evidence synthesis. Rigorous peer review and editorial oversight.

Theory

Meta-analysis of 38 randomised controlled trials investigating mindfulness-based interventions for chronic pain conditions. Uses established frameworks from pain neuroscience and psychological interventions.

Methods & Data

Systematic review following Cochrane protocols. Included 3,500+ participants across multiple pain conditions. Preregistered methodology. Independent reviewers assessed bias. Full PRISMA checklist provided. **Data availability:** All extracted data and analysis code publicly available.

Conclusion

Cautious and specific. Reports a small-to-moderate effect size for pain reduction (d=0.33) but notes high heterogeneity between studies. Acknowledges mindfulness works better for some pain types than others and cannot replace medical treatment.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study E · Fictional study

TITLE

The predictive relationship between income inequality and population health outcomes: A longitudinal study

Access

Open Access.

Source

Top-Tier Public Health Journal. Known for rigorous statistical review and replication-friendly practices.

Theory

Uses 40 years of data across 50 countries to examine Gini coefficient and health metrics. Builds on established social epidemiology theories and addresses confounders like absolute poverty and healthcare spending.

Methods & Data

Sophisticated multi-level modelling using WHO and World Bank datasets. Preregistered analysis plan. Extensive sensitivity analyses. Controls for GDP, healthcare access, education. **Data availability:** All code and datasets publicly archived. Note: approximately 15% of country-year observations had missing health data requiring multiple imputation; sensitivity analyses confirm findings are robust, but the imputation assumptions are consequential.

Conclusion

Robust but carefully worded. Shows strong association between inequality and life expectancy, infant mortality and mental health even after controlling for absolute wealth. Acknowledges causation cannot be proven from observational data.

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

Study F · Fictional study

TITLE

Correction of the sickle cell mutation in a human patient trial using CRISPR gene editing

Access

Behind paywall.

Source

Top-Tier Medical Journal. Requires independent data monitoring boards for clinical trials.

Theory

Uses CRISPR-Cas9 to edit patient's own haematopoietic stem cells, correcting the genetic mutation causing sickle cell disease. Grounded in molecular biology and gene therapy.

Methods & Data

Single-arm trial (no control group; common in early gene-therapy trials, but it makes the benefit harder to judge). *N*=15 patients with severe sickle cell disease. 2-year follow-up. Rigorous safety monitoring. **Data availability:** Patient-level data restricted; summary data in paper.

Conclusion

The title overclaims for a 15-patient single-arm trial. 'Correction of the mutation' implies completeness that two years of follow-up cannot yet establish. The paper itself is more careful: all patients showed meaningful improvement, but long-term durability is unknown, the procedure is extremely expensive and off-target editing was detected in 2 patients (ongoing monitoring required).

Your answers

1. Predict the crowd

What will the average rating from all workshop participants be? Your score compares your prediction with a target that also counts the expert's rating heavily.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

2. Your own rating

How good is this study overall? Weigh up the six criteria.

Very low (1) (2) (3) (4) (5) (6) (7) Very high

For the facilitator

Answer key

The expert's feedback is the same as the app shows after each study.

Study	Title	Expert rating	Quality	Dashboard number
Study A	The neuroprotective effects of a daily herbal tea infusion on Alzheimer's disease prevention	1	Low quality	4
Study B	A causal link between Instagram use and adolescent anxiety	4	Medium quality	6
Study C	An evolutionary explanation for the correlation between height and CEO status in males	2	Low quality	11
Study D	The efficacy of mindfulness meditation as an intervention for chronic pain: A systematic review	7	High quality	24
Study E	The predictive relationship between income inequality and population health outcomes: A longitudinal study	5	Medium quality	27
Study F	Correction of the sickle cell mutation in a human patient trial using CRISPR gene editing	6	High quality	43

Answer key

Study A: The neuroprotective effects of a daily herbal tea infusion on Alzheimer's disease prevention

Expert rating: 1 out of 7 · Low quality

Weak on every criterion except access. It appeared in a predatory journal with no real peer review and ignores what is known about Alzheimer's disease. For six weeks it followed 40 people who already drank the tea, with no proper control group, yet it claims long-term prevention. Free access cannot make up for that: it gets the lowest rating.

Criterion by criterion

Criterion	Band	Note
Title	Low	Words like 'neuroprotective' (protecting brain cells) and 'prevention' promise far more than a six-week study of 40 tea drinkers could ever show.
Access	High	Anyone can read it free online. Many open-access journals charge authors rather than readers, so the fee is not itself a warning sign.
Source	Low	A predatory journal: its website is in broken English and promises 'publication within 48 hours for a fee', showing it has no peer review (checks by independent experts).
Theory	Low	Loosely based on folk medicine, it ignores amyloid plaques and tau tangles, the protein build-ups in the brain that are the main biological signs of Alzheimer's.
Methods & Data	Low	Forty people already drinking the tea were followed for six weeks with no randomly chosen comparison group. Brief phone calls replaced standard in-person thinking tests; raw data are withheld.
Conclusion	Low	It treats a short-term feeling of alertness, likely a placebo effect (feeling better because you expect to), as proof of long-term protection against a complex disease.

Answer key

Study B: A causal link between Instagram use and adolescent anxiety

Expert rating: 4 out of 7 · Medium quality

Standard theory, a large sample and a legitimate peer-reviewed journal give this study a fair base. Its title claims cause and effect, yet a one-off survey of what teenagers said about themselves can only show a link. It also ignores whether anxious teenagers simply turn to social media more. That leaves it in the middle of the scale.

Criterion by criterion

Criterion	Band	Note
Title	Low	It announces a 'causal link', yet a survey taken at a single moment can only show that the two go together: correlation is not causation.
Access	Medium	Readers need a university library subscription, so most members of the public cannot read the full paper.
Source	Medium	A legitimate mid-tier peer-reviewed journal, but one that sometimes publishes studies with catchy titles to boost its impact factor (a measure of how often its papers are cited).
Theory	Medium	It uses standard ideas about social comparison (measuring yourself against others) and self-esteem but assumes Instagram directly causes anxiety, overlooking how complex teenage mental health is.
Methods & Data	Medium	Surveying 5,000 teenagers gives a large sample, but both hours of use and anxiety were self-reported at a single point in time and raw data are only available on request.
Conclusion	Medium	The strong link it reports fits the data, but the authors hint at cause and effect and never ask whether anxious teenagers simply turn to social media more.

Answer key

Study C: An evolutionary explanation for the correlation between height and CEO status in males

Expert rating: 2 out of 7 · Low quality

This free preprint, not yet checked by independent experts, links height with becoming a CEO (company boss), then explains the link with an evolutionary story it never tests. It ignores the simpler explanation that people tend to promote tall men. These flaws in theory and conclusion weigh far more than its easy access and reasonable comparison.

Criterion by criterion

Criterion	Band	Note
Title	Medium	It honestly calls the height link a correlation, but offering ‘an evolutionary explanation’ presents an untested story as if the data supported it.
Access	High	Anyone can download it free from a preprint server (a public site for sharing papers before review), though being free to read says nothing about quality.
Source	Medium	As a preprint it has not yet been checked by independent experts (peer review) or accepted by any journal, so its claims remain unverified.
Theory	Low	It treats modern firms like prehistoric hunter-gatherer groups and ignores the ‘halo effect’, a social bias in which one admired trait such as height colours our whole judgement.
Methods & Data	Medium	Comparing CEOs’ heights at the 500 largest US firms with the national average can show a link, but nothing measured leadership or performance and data are only on request.
Conclusion	Low	It mistakes a social bias (we tend to promote tall men) for an inborn ability (tall men are better leaders), treating a correlation as if it proved a cause.

Answer key

Study D: The efficacy of mindfulness meditation as an intervention for chronic pain: A systematic review

Expert rating: 7 out of 7 · High quality

This review is strong on every criterion. It is free to read, published by Cochrane and pools 38 randomised trials with more than 3,500 participants. Its plan was set in advance, its data and code are open and it honestly reports a modest benefit that varies between trials and pain types. That earns it the top score.

Criterion by criterion

Criterion	Band	Note
Title	High	Plain and accurate: it names the question and says this is a systematic review (a study that gathers and combines all suitable earlier trials), with no promise of a cure.
Access	High	Free to read through the Cochrane Library, so anyone can check the full review and how it was done.
Source	High	The Cochrane database is a gold-standard source for summing up research evidence, with rigorous peer review (checks by other experts) and careful editorial oversight.
Theory	High	Pooling 38 randomised controlled trials (where people are put into groups by chance) and drawing on established pain science, it uses earlier research carefully to answer one clear question.
Methods & Data	High	Covering more than 3,500 participants, it followed Cochrane rules and a plan published in advance, had independent reviewers check the trials for bias and shares all data and code.
Conclusion	High	It reports a small-to-moderate reduction in pain, notes that results varied widely between trials and says mindfulness suits some pain types better and cannot replace medical treatment.

Answer key

Study E: The predictive relationship between income inequality and population health outcomes: A longitudinal study

Expert rating: 5 out of 7 · Medium quality

Strong on every criterion: open access, a top public health journal, a plan fixed in advance, shared code and data and a careful conclusion. It still gets only 5 because the expert weighs two limits heavily: data from observation, not experiment, cannot prove that inequality harms health and about 15% of the health figures had to be estimated.

Criterion by criterion

Criterion	Band	Note
Title	High	'Predictive relationship' claims a link rather than proof of cause and 'longitudinal' (following the same countries over many years) describes the design exactly.
Access	High	It is open access, so anyone can read the full paper free of charge and check the work for themselves.
Source	High	A top public health journal known for careful statistical review and for practices that help other researchers repeat (replicate) published studies.
Theory	High	Using 40 years of data from 50 countries, it builds on established social epidemiology (how society shapes health) and accounts for rival explanations such as absolute poverty.
Methods & Data	High	The analysis plan was recorded in advance (preregistered) and all code and data are public, but about 15% of yearly country records had missing health data filled in by estimation.
Conclusion	High	It finds a strong link between inequality and life expectancy, infant mortality and mental health, yet states plainly that observational data (gathered without an experiment) cannot prove cause.

Answer key

Study F: Correction of the sickle cell mutation in a human patient trial using CRISPR gene editing

Expert rating: 6 out of 7 · High quality

A gene-editing trial for sickle cell disease in a top medical journal, with close safety monitoring and an honest paper that reports unintended edits in two patients. The expert's 6 gives most weight to these strengths; on the rubric alone, the paywall, just 15 patients with no comparison group and an overclaiming title would put it nearer the middle.

Criterion by criterion

Criterion	Band	Note
Title	Medium	It names the method and makes clear this was a patient trial, but 'correction' suggests a complete, lasting fix that a two-year trial of 15 patients cannot yet show.
Access	Low	A paywall stops most readers from checking the trial for themselves, although the price of a paper says nothing about whether the research is sound.
Source	High	A top medical journal that insists trials are overseen by an independent data monitoring board (outside experts who check safety and results while the trial runs).
Theory	High	It builds on molecular biology and gene therapy, using CRISPR (a tool for cutting and editing DNA) on patients' own blood-forming stem cells, aiming to correct the faulty gene.
Methods & Data	Low	Just 15 patients and no comparison group make the benefit harder to judge and patients' individual data stay private, although close safety monitoring over two years is a strength.
Conclusion	High	Unlike the title, the paper is careful: all patients improved, but it admits nobody knows how long this lasts, notes the high cost and reports unintended edits in two patients.

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